

EBN122 CLASS TEST 3 (31 OCT 2014)

Vraag 1 [2]

Die spanning oor 'n 1 mH induktor is $v_L(t) = 2e^{-2t}$ mV. Aanvaar $i_L(0) = 0$ A en bepaal $i_L(t)$ wanneer $t = 0.25$ s.

Question 1 [2]

The voltage across a 1 mH inductor is $v_L(t) = 2e^{-2t}$ mV. Assume $i_L(0) = 0$ A and determine $i_L(t)$ when $t = 0.25$ s.

Handwritten calculations for Question 1:

$$v = L \frac{di}{dt}$$

$$2e^{-2t} = 1 \times 10^{-3} \frac{di}{dt}$$

$$2e^{-2t} \times 10^3 = \frac{di}{dt}$$

$$2000e^{-2t} = \frac{di}{dt}$$

$$L \int v dt$$

$$1 \times 10^{-3} \int 2000e^{-2t} dt$$

$$2e^{-2t} = 2e^{-2 \times 0.25} = 2e^{-0.5}$$

$$2e^{-0.5} = 2 \times 0.6065 = 1.213$$

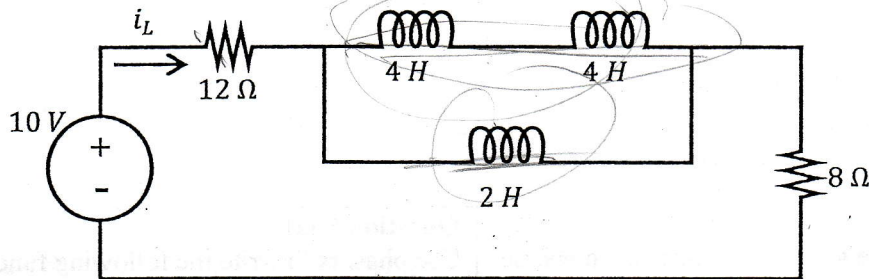
$$i_L(t) = 1.213 \text{ A}$$

Vraag 2 [3]

Bepaal die ekwivalente induktansie in die kringbaan. Bepaal i_L en die totale energie gestoor in die induktore.

Question 2 [3]

Determine the equivalent inductance in the circuit. Determine i_L and the total energy stored in the inductors.



Handwritten calculations for Question 2:

Equivalent inductance:

$$4H + 4H = 8H$$

$$8H \parallel 2H = \frac{8 \times 2}{8 + 2} = 1.6H$$

Current i_L :

$$I_1 = 20 \Omega$$

$$V = IR$$

$$10V = I \times 20 \Omega$$

$$I = \frac{10}{20} = 0.5A$$

Total energy stored:

$$W = \frac{1}{2} LI^2$$

$$W = \frac{1}{2} \times 1.6 \times (0.5)^2$$

$$W = 0.2J$$

Vraag 3 [2]

Watter sinusied is nalopend en met hoeveel grade?

$$v_1(t) = 2 \cos(\omega t - 100^\circ) \text{ V}$$

$$v_2(t) = -10 \sin(\omega t + 30^\circ) \text{ V}$$

Question 3 [2]

Which sinusoid is lagging and by how many degrees?

$$v_1(t) = 2 \cos(\omega t - 100^\circ) \text{ V}$$

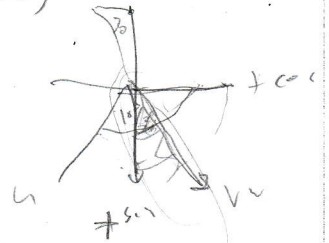
$$v_2(t) = -10 \sin(\omega t + 30^\circ) \text{ V}$$

$$2 \cos(\omega t - 100^\circ)$$

$$2 \angle -100$$

$$10 \cos(\omega t - 60^\circ)$$

$$10 \angle -60$$

**Vraag 4 [2]**

Bereken en skryf die antwoord in reghoekige formaat:

$$5(2 - j2\sqrt{3})^{1/2} + (3\sqrt{3} + 2j)(-5 - j\sqrt{3})^*$$

Question 4 [2]

Calculate and write the answer in rectangular form:

$$5(2 - j2\sqrt{3})^{1/2} + (3\sqrt{3} + 2j)(-5 - j\sqrt{3})^*$$

$$5 \sqrt{2 - j2\sqrt{3}} +$$

$$5 \sqrt{4 \angle -60} +$$

$$5(2 \angle -30) + (5.567 \angle 21.05) (5.291 \angle +160.893)$$

$$(\sqrt{3} - j)$$

$$+ 29.45 \angle 181.94$$

$$+ -29.43 - j - 20.77 - j$$

Vraag 5 [2]

Gebruik fasors en skryf die volgende funksie as 'n enkele cosinus uitdrukking:

$$-8\cos(\omega t + 120^\circ) - 6\sin(\omega t - 120^\circ)$$

Die antwoord moet in die volgende vorm wees:

$$A\cos(\omega t + \phi), A > 0, -180^\circ \leq \phi \leq 180^\circ$$

Question 5 [2]

Use phasors to write the following function as a single cosine expression:

$$-8\cos(\omega t + 120^\circ) - 6\sin(\omega t - 120^\circ)$$

The answer must have the following form:

$$A\cos(\omega t + \phi), A > 0, -180^\circ \leq \phi \leq 180^\circ$$

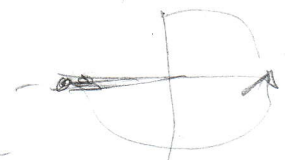
$$\frac{1}{2}$$

$$(86 + 4\sqrt{3}j) +$$

$$-8 \angle 120^\circ + \cos -210$$

$$-(8 \angle 120) + (6 \angle -210)$$

$$8 \angle 60$$



$$86 + 4\sqrt{3} + 3\sqrt{3} + 5j + 4 + 6.92j - 5.19 + 3j$$

Vraag 6 [2]

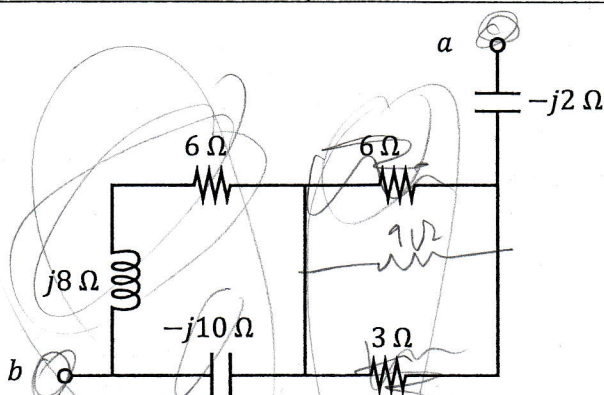
Bepaal die ekwivalente impedansie tussen terminale a en b vir onderstaande kringbaan.

(Skryf die antwoord in reghoekige formaat.)

Question 6 [2]

Determine the equivalent impedance between terminals a and b for the circuit below.

(Write the answer in rectangular format.)



$$9 \rightarrow \frac{(6 + j8)(-j10)}{6 + j8 + 10j} = -2$$

$$24 - 5j$$

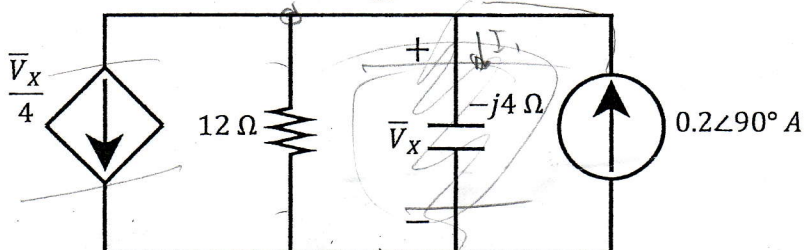
$$-7$$

Vraag 7 [2]

Bepaal $\bar{V}_x$. Skryf antwoord in polêre formaat.

Question 7 [2]

Determine $\bar{V}_x$. Write answer in polar format.



$$\frac{V_x}{12} = \frac{V_x}{-j4} - \frac{V_x}{12} - \frac{V_x}{4}$$

$$0.2 \angle 90^\circ = \frac{V_x}{-j4} + \frac{V_x}{12} + \frac{V_x}{4}$$

$$V_x \left(\frac{1}{-j4} + \frac{1}{12} + \frac{1}{4} \right) = \frac{V_x - 3V_x}{12}$$

$$0.2 \angle 90^\circ + \frac{2V_x}{12} = 0$$

$$0.2 \angle 90^\circ = V_x \left(\frac{1}{3} + \frac{1}{4j} \right)$$

$$\frac{1}{j} \angle 90^\circ = V_x (0.4166 \angle 36.866^\circ)$$